Natural Language Processing - Quiz 2

25 Marks , Time: 30 minutes

Name:____________ **Roll no.:**____________ **Date:**17.02.2026

Instructions: Write all answers on the back-side of the page. Best of luck!

1 MCQ 10X1.5=15, for wrong choice, penalty is 0.5

Q1: Suppose you have a term–document matrix with N documents. A term t appears in exactly 1% of the documents. Which of the following is closest to its inverse document frequency idf(t)=$log_{10}(\frac{N}{df_t})$?

(a) 0
(b) 1
(c) 2
(d) Can't be computed based on given information

Q2: You compute cosine similarity between three word vectors in a 3-dimensional space:
v_X=(3,0,4)
v_Y=(6,0,8)
v_Z =(0,5,0)

Which statement is correct?

(a) cos(X,Y)=1 and cos(X,Z)=0
(b) cos(X,Y)=0.5 and cos(X,Z)=0
(c) cos(X,Y)=1 and cos(Y,Z)=0.5
(d) All pairwise cosine similarities are strictly between 0 and 1

Q3: Skip-gram with negative sampling (SGNS) trains a classifier that, for each target–context pair (t,c), predicts whether the pair is sampled from real text or from a noise distribution. Which of the following best describes what becomes the word embedding?

(a) The bias term of the logistic regression classifier
(b) The gradient of the loss for each training example
(c) The probability assigned to real pairs after training
(d) The learned weight vectors associated with targets and/or contexts

Q4: In word2vec's SGNS, increasing the number of negative samples per positive example (k) typically has which effect, holding everything else fixed?

(a) It reduces training time but harms embedding quality
(b) It increases training time but can improve the quality of the learned space
(c) It has no effect as long as k $\geq$ 1
(d) It improves the quality with same amount of training time

Q5: Consider two words, "doctor" and "nurse", in a static embedding space that was trained on a corpus with strong gender stereotypes. Their nearest neighbors show that "doctor" is close to "he" and "nurse" is close to "she". Which statement is most accurate?

(a) This proves that the model has eliminated gender bias by separating occupations
(b) The embeddings encode historical gender stereotypes present in the training data
(c) The bias arises purely from random initialization
(d) Cosine similarity is invariant to such biases

Q6: In a simple RNN with recurrence h_t =g(Uh_{t-1} +Wx_t), which statement about inference is correct?

(a) Computing h_t requires h_{t-1}, so inference proceeds incrementally from t=1 to t=n.
(b) All hidden states for a sequence can be computed in parallel because weights are shared.
(c) Inference starts from the last token and proceeds backward.
(d) The hidden state at time t is independent of the previous inputs once W is known.

Q7: For an RNN language model, which of the following best distinguishes it from an n-gram language model?

(a) It always uses a fixed context window

of size 3.
(b) It conditions on a one-hot vector of the next word rather than the previous words.
(c) It has no fixed context size; h_{t-1} can, in principle, summarize the entire history.
(d) It cannot be trained with cross-entropy loss.

Q8: In training an RNN language model, teacher forcing refers to:
(a) Forcing the model to predict the next word without using any previous words.
(b) Always feeding the model its own predicted next word as the next input.
(c) Training only on sequences where the model already achieves low loss.
(d) Always giving the model the gold previous word as input when predicting the next word.

Q9: What main problem do LSTMs address compared to simple RNNs?
(a) The inability of RNNs to process variable-length sequences.
(b) The high memory cost of storing all hidden states during backpropagation.
(c) The vanishing (and exploding) gradient problem when modeling long-distance dependencies.
(d) The fact that RNNs cannot be stacked or used bidirectionally.

Q10: Interpolation differs from backoff because interpolation:
(a) Uses only lower-order models
(b) Discards higher-order probabilities
(c) Combines multiple N-gram models simultaneously
(d) Assigns zero probability to unseen events

2 Fill in the Blanks

(Q11) In tf-idf weighting, the inverse document frequency term is based on the ratio N/df_t , where N is the total number of documents and df_t is the number of documents containing term t; this gives higher weight to terms that appear in ______________________ documents. [1]

(Q12) In Osgood et al. (1957), connotation is modeled as a point in a 3-dimensional affective space whose axes are ______________________, ______________________, and ______________________. [1]

(Q13) The **ReLU** activation function is defined as $y = \max(z, 0)$; its derivative with respect to z is 0 for $z < 0$ and _______ for $z \geq 0$.
[1]

3 Numerical

(a) Consider a simple RNN with scalar inputs, hidden states, and outputs. The recurrence is

$$h_t = \tanh(Uh_{t-1}+Wx_t), \qquad \hat{y}_t = \text{softmax}(Vh_t).$$

Assume:

- Vocabulary size $|V| = 3$, so $\hat{y}_t$ is a 3-dimensional probability vector.
- At time $t = 1$: input one-hot $x_1 = (1, 0, 0)^\top$, and you may treat the scalar pre-activation as $Wx_1 = 0.5$.
- Parameters: $U = 0.4$, $W = 0.5$, $V = [1.0, \ -0.5, \ 0.0]$.
- Initial hidden state: $h_0 = 0$.

(a) Compute h_1.
(b) Compute the unnormalized logits $z = Vh_1$.
(c) Compute the softmax probabilities $\hat{y}_1$.

[**2+2+3=7 Marks**]
(a) h1 =
(b) z =
(c) $\hat{y}_1$=

Answer **(a) Hidden state h_1.**
The pre-activation for the hidden state is

$$a_1 = Uh_0 + Wx_1 = 0.4 \cdot 0 + 0.5 = 0.5.$$

Hence

$$h_1 = \tanh(0.5).$$

Numerically, $\tanh(0.5) \approx 0.4621$, so

$$h_1 \approx 0.4621.$$

(b) Logits z.
The logits are

$$z = Vh_1 = \begin{bmatrix} 1.0 \\ -0.5 \\ 0.0 \end{bmatrix} h_1 = \begin{bmatrix} 1.0h_1 \\ -0.5h_1 \\ 0 \end{bmatrix} \approx \begin{bmatrix} 0.4621 \\ -0.2310 \\ 0 \end{bmatrix}.$$

(c) Softmax probabilities $\hat{y}_1$.
We compute

$$\hat{y}_{1,i} = \frac{\exp(z_i)}{\sum_{j=1}^{3} \exp(z_j)}.$$

Approximate exponentials:

$$\exp(0.4621) \approx 1.587, \quad \exp(-0.2310) \approx 0.794, \quad \exp(0) = 1.$$

Denominator:

$$Z = 1.587 + 0.794 + 1 = 3.381.$$

Thus

$$\hat{y}_{1,1} = \frac{1.587}{3.381} \approx 0.469, \quad \hat{y}_{1,2} = \frac{0.794}{3.381} \approx 0.235, \quad \hat{y}_{1,3} = \frac{1}{3.381} \approx 0.296.$$

So

$$\hat{y}_1 \approx [0.469,\ 0.235,\ 0.296].$$

Answers

Q1: c

Q2: a

Q3: d

Q4: b

Q5: b

Q6: a

Q7: c

Q8: d

Q9: c

Q10: c

Q11: fewer

Q12: valence, arousal, dominance [order does not matter]

Q13: 1